

Shopping goal specificity matters: How algorithmic recommendations influence adoption intention

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ABSTRACT

In the age of online shopping, consumers frequently engage with shopping apps in hybrid modes, switching between goal-specific searches and serendipitous browsing. This behavior poses a pressing challenge for algorithm-driven recommendation systems. While existing studies prioritize algorithmic accuracy, they largely neglect the dynamic congruence between recommendations (content-based vs. user-based) and consumers' shifting goal specificity. This study addresses the aforementioned gap by focusing on adaptive algorithmic recommendation strategies. Drawing on the information processing framework of the information adoption model with the decision-making mechanism in shopping goals theory, we examine the impact of the two predominant algorithmic recommendations (content-based and user-based) on recommendation adoption intention. Our hypotheses are verified by the results of three 2×2 between-subjects experiments. The findings reveal that content-based (user-based) recommendations are more effective when consumers have high (low) specific goals, and this effectiveness is due to the fact that matching the two enhances consumers' perceived diagnosticity (serendipity). Furthermore, the interaction between shopping goal specificity and algorithmic recommendations is moderated by shopping motivation. Building on shopping goals theory and information adoption model, these findings advance recommendation systems literature by establishing goal specificity as a critical moderator and introducing dual cognitive pathways in information adoption. This insight not only resolves the efficiency of recommendation systems but also provides actionable insights for segment-specific targeting in online shopping.

1. Introduction

Imagine that one day; while browsing online, you open a shopping app (e.g., Amazon). Your goal might be to mindlessly surf, or it might be to purchase a specific item. As you do, you may encounter an interface labeled "Guess your favorite products" or "Products liked by users similar to you". In such a case, would you adopt the recommended products?

A global shift in shopping behavior has taken place over the past decade, marked by the rapid rise of mobile app-based shopping (Forbes, 2017). Increasingly, consumers are browsing e-commerce platforms without a clearly articulated or specific shopping goal (Patel et al., 2020). However, the vast array of items can result in cognitive burden, making it difficult for users to identify preferred items (Broniarczyk and Griffin, 2014; Z. Sun et al., 2019). The challenge is reflected in the fact that 43.8 % of online shoppers browse without adding any items to their virtual carts (Finance Oline, 2021), highlighting the limitations of

current product recommendation strategies. This suggests a misalignment between existing recommendation methods and consumer needs, underscoring the importance of exploring new solutions. Automatic recommendation systems have the potential to enhance sales (Bodapati, 2008), thereby offering promising directions. As a cutting-edge automatic recommendation system, algorithmic recommendations hold significant potential to enhance recommendation effectiveness by mitigating information overload (Elhajjar, 2024).

Algorithmic recommendations are automated, data-driven systems widely used to deliver personalized suggestions to consumers (Gai and Klesse, 2019). Contemporary techniques primarily include content-based recommendations, user-based recommendations, and hybrid methods that combine both (Z. Sun et al., 2019). Content-based recommendations analyze browsing history and choices to predict suitable items (Chinchanachokchai et al., 2021). In contrast, user-based recommendations suggest items favored by one user to others with similar preferences (Schafer et al., 2007). This method mirrors

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real-world social purchasing behavior, in which consumers make decisions based on peer recommendations (Yan et al., 2016). Researchers generally agree that content-based and user-based recommendations offer distinct strengths. Specifically, content-based recommendations tend to outperform user-based ones in terms of similarity, whereas user-based recommendations offer greater diversity (Ochi et al., 2010). Past research has demonstrated the efficacy of algorithmic recommendations as intelligent service technologies (Banker and Khetani, 2019; Y. Jiang et al., 2015), the issue of “on what basis should recommendations adjust their underlying algorithmic logic” remains largely unexplored.

According to the Shopping Goals Theory, the shopping goals of consumers become progressively more specific (L. Lee and Ariely, 2006). Building on this perspective, research has introduced *goal specificity*, characterized as the lack of clarity or imprecision surrounding the precise targets or outcomes intended to be achieved (Sarantopoulos et al., 2019). This concept underpins the recommendation logic proposed in this study. While prior research has highlighted the importance of aligning shopping goal specificity with selling strategies in physical retail environments (Y. Kim and McFarland, 2024), the distinctive characteristics (e.g., personalization and disorganized information) of mobile shopping (Guo et al., 2017) complicate the direct application of these findings to digital contexts. The Information Adoption Model offers a suitable analytical framework to examine this issue. According to the IAM, individuals are influenced by information through a central path which prioritizes argument quality, and a peripheral path which prioritizes source credibility (Sussman and Siegal, 2003). By integrating shopping goals theory with IAM, we propose that consumers with highly specific goals are more likely to engage in central processing (argument quality), whereas consumers with less specific goals are more inclined to rely on peripheral cues (source credibility). Accordingly, we hypothesize that user-based recommendations are more effective for consumers with low shopping goal specificity, as these recommendations leverage the preferences of a broad user base and thus offer high source credibility (Darani and Aghaie, 2025). In contrast, content-based recommendations are more effective for consumers with high goal specificity, as they provide tailored information aligned with consumers' concrete goals and thus offer high argument quality (Fujita et al., 2007). This theoretical framework represents a key contribution of this study, extending prior research on both IAM and shopping goals theory. This insight also yields managerial implications, such as enabling platforms to tailor product suggestions based on whether users are “just browsing” or “looking for something specific.”

Another significant contribution of this study lies in identifying an intrinsic psychological mechanism that underpins the above matching effect. Recent research suggests that effective recommendation systems should account for not only diagnosticity (e.g. consumers feel that a comparison list of each product's price, features, advantages, etc. can facilitate their decision-making process) but also serendipity (e.g. consumers experience surprise and interest when they buy mystery box) to enhance user engagement (Kwon et al., 2021). Importantly, perceived diagnosticity and perceived serendipity are not mutually exclusive; rather, they represent distinct yet complementary dimensions of the consumer shopping experience (Yi et al., 2017). While past research has recognized the significance of similarity and serendipity, they often conflate their mediating roles within a unified explanatory framework (Yang et al., 2023a, 2023b). Departing from this tradition, we propose a dual-path mediation model: perceived diagnosticity mediates goal-driven matching (e.g., content-based recommendations for consumers with high shopping goal specificity), whereas perceived serendipity mediates exploration-oriented matching (e.g., user-based recommendations for consumers with low shopping goal specificity).

The effectiveness of algorithmic recommendations is influenced not only by consumers' shopping goal specificity (what they intend to buy) but also by the underlying motivations driving those purchases (why they intend to buy) (Y. Kim and McFarland, 2024). Past research has identified hedonic and utilitarian motives as the primary shopping

purposes of consumers (Babin et al., 1994). Utilitarian motivations are associated with goal-oriented and efficiency-driven behavior (S. Lee et al., 2025). This study suggests that such motivations will activate central processing routes, thereby enhancing the effectiveness of content-based recommendations for consumers with high shopping goal specificity. In contrast, hedonic motivations are linked to experiential and exploration-oriented behavior (S. Kim and Eastin, 2011). These motivations are posited to trigger peripheral processing routes, which support the effectiveness of user-based recommendations for consumers with low shopping goal specificity. This conceptualization serves as a further extension of the IAM, highlighting how shopping motivations moderate the pathway through which information is processed. By integrating the ‘goal-motivation-algorithm’ framework, this research offers granular insights into refinement strategies to optimize recommendation strategies.

The rest of this research is structured as follows. Section 2 offers an overview of the theoretical background and outlines the development of the hypotheses. Section 3 details the methodology employed and presents the key results, followed by an in-depth discussion. Finally, Section 4 concludes the research, discussing the theoretical and managerial implications of the findings and suggesting potential issues for future research.

2. Theoretical background

2.1. Information adoption model

The Information Adoption Model is proposed by Sussman and Siegal (2003), which suggests that individuals evaluate persuasive information through two separate pathways. One of them focuses on the argument quality (central route) and the other focuses on the source credibility (peripheral route) (Erkan and Evans, 2016). *Argument quality* refers to the perception of information's completeness, consistency, and accuracy (Y. Sun et al., 2019), whereas *source credibility* pertains to perceptions of the source's believability, competence, and trustworthiness (Gunawan and Huarng, 2015). According to the model, these two routes jointly influence the perceived usefulness, which in turn affects individuals' willingness to adopt it (Shen et al., 2016). The IAM has since been extended and validated across various contexts, demonstrating its robustness as a framework for understanding information adoption. Consumers are exposed to vast volumes of recommendation information, the IAM provides a suitable analytical lens for examining how users evaluate and respond to such information. Based on IAM, we posit that the interaction between algorithm recommendations and shopping goal specificity exerts a significant influence on consumer adoption intentions, operating through the mechanisms of perceived serendipity and diagnosticity, respectively. The framework of this research is shown in Fig. 1.

2.2. Algorithmic recommendations

Over the past two decades, online retailers have thrived alongside the expansion of e-commerce. Amazon, for instance, reported \$386 billion in revenue in 2020, marking a 38 % increase from the previous year, with an annual growth exceeding \$100 billion (Amazon, 2021). An increasing number of e-commerce platforms are leveraging algorithmic recommendations to drive online sales. Algorithmic recommendations are based on consumers' shopping history and search behaviors, and even use eye movement analysis (Chang and Park, 2024; Jung et al., 2013) to estimate consumer preferences. In order to better target consumers, some clustering algorithms such as GA K-means can be used to segment the market (K. Kim and Ahn, 2008). A body of research has previously investigated the phenomena of algorithm aversion (Dietvorst et al., 2015) and algorithm appreciation, indicating that AI (Human) recommender exhibits optimal performance when confronted with fact-based (affect-based) information (S. Li et al., 2025; Longoni and

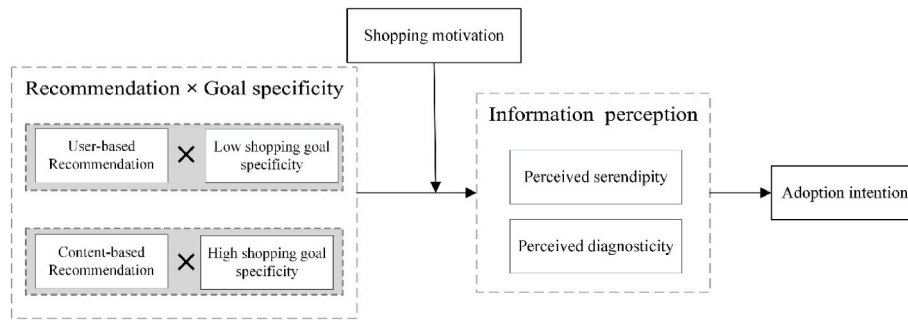


Fig. 1. Research framework.

Cian, 2022). This study focuses on comparing different algorithmic recommendation methods, as different algorithms can also output more factual/affective information. Algorithmic recommendations are composed of content-based recommendations, user-based recommendations, and a hybrid of the two (Z. Sun et al., 2019). Based on extant studies (Choi et al., 2012; Yan et al., 2016; Zhang et al., 2022), this study proposes content-based and user-based recommendations and offers a comparative analysis of their respective recommendation effects. Content-based recommendation is a type of recommendation system that identifies products with similar attributes (Chinchanchokchai et al., 2021). These recommendations are generated based on consumers' preferences for specific product types or attributes (S. Huang, 2011). User-based recommendation is another type of recommendation system that focuses on identifying consumers who are similar in terms of behavioral characteristics and product preferences (Köhler et al., 2016). It recommends products preferred by consumers exhibiting similar preferences or behaviors (Darani and Aghaie, 2025).

The comparative effectiveness of different algorithmic recommendation methods has attracted considerable scholarly attention. Arguments have been made that user-based recommendations outperform content-based recommendations due to the inclusion of multiple pieces of information (Gai and Klesse, 2019; Zhang et al., 2022). Some studies also point out that the relative efficiency of user-based and content-based recommendations is shaped by various factors, which include consumer expertise (Chinchanchokchai et al., 2021), algorithm designs (Darani and Aghaie, 2025), as well as the number of recommendations (Zhang et al., 2022). However, existing studies have overlooked the significance of shopping goals (Bell et al., 2011) as the beginning of the shopping journey. Research in brick-and-mortar retail environments has underscored that aligning sales influence strategies with shopping goal specificity can enhance positive outcomes (Y. Kim and McFarland, 2024). Similarly, this research suggests that algorithmic recommendations (sales influence strategies in e-commerce) should be aligned with consumers' shopping goal specificity to ensure optimal outcomes.

2.3. Matching algorithmic recommendations with shopping goal specificity

The shopping goals theory posits that consumers' shopping journey can be categorized into two stages: the initial phase and the purchase phase (L. Lee and Ariely, 2006). In the initial stage, consumers adopt a more abstract way of thinking and form a choice set from a number of items (Yi et al., 2023). At the second stage, consumers adopt a more concrete way of thinking, and make a comparison (Fujita et al., 2007) and choose the final purchased item from the choice set. In accordance with this theory, scholars have determined that the acceptance of products recommended by algorithms varies across different stages (Yan et al., 2016). This acceptance is also influenced by decision-making styles, such as maximizers versus satisficers (Virdi et al., 2020). However, these studies have overlooked the possibility that consumers may

transition between specific and vague goals during the shopping process. In light of this perspective, certain scholars have advanced the concept of goal specificity (Sarantopoulos et al., 2019). Consumers may open mobile shopping apps to look for something specific or just look around. A more specific goal is defined as high shopping goal specificity (high SGS), such as a consumer using a shopping app to purchase potato chips. In contrast, a vague goal is considered as low shopping goal specificity (low SGS), as exemplified by a consumer browsing the app to just look around (Y. Kim and McFarland, 2024). Thus, it is necessary to tailor recommendations to consumers based on the specific purpose for which they use the shopping apps (L. Chen et al., 2019). When consumers have defined goals, relevant recommendation content that aligns with these goals can facilitate adoption (Wang et al., 2022). This study posits that content-based (user-based) recommendations are more effective when consumers have high (low) SGS.

Consumers with low SGS frequently engage with shopping apps to pass the time. These consumers are generally more receptive to diverse product information and tend to prefer recommendations that incorporate a broader range of attributes (Ochi et al., 2010). According to the IAM, these consumers demonstrate a higher propensity to process information through peripheral routes and prioritize source credibility (Sussman and Siegal, 2003). User-based recommendations suggest products favored by similar customers. These recommendations have the potential to promote consumer imitation (Zhang et al., 2022) and enhance recommendation credibility (Xu et al., 2023). Thereby, they can increase consumers' willingness to adopt the recommendation. Additionally, this method offers consumers a more extensive array of choices, thereby facilitating a more serendipitous experience (P. Li and Tuzhilin, 2024). Bharadwaj (2022) offers compelling evidence that visually mediated stimuli can evoke strong affective responses, leading to impulsive purchase intentions, particularly among emotionally vulnerable youth consumers. This aligns with our proposition that algorithmic recommendations - especially visually rich or emotionally charged ones - are more persuasive under low goal specificity, where affect dominates cognition. The parallels between the simulated ad environment and algorithmic visual presentation support the notion that affective visual stimuli serve as powerful decision catalysts in ambiguous shopping contexts (Bharadwaj, 2022). In contrast, content-based recommendations, which aim to present similar items, may raise concerns about opportunistic behavior by sellers (Gallin and Portes, 2024) or induce consumer fatigue due to repetitive suggestions. Consequently, this strategy may weaken source credibility and reduce the effectiveness of the recommendation under peripheral processing.

Consumers with high SGS frequently engage with shopping apps to search for something specific. According to the IAM, these consumers prioritize the quality of the arguments and carefully scrutinize product details when they make decisions (Song et al., 2017). The argument quality is determined by the perception of information's completeness, consistency, and accuracy (Y. Sun et al., 2019). Compared to user-based recommendations, content-based recommendations offer greater consistency and accuracy (Ochi et al., 2010), better aligning with

consumers' preferences (Virdi et al., 2020). This information is important to consumers' evaluation of product attributes, particularly under central processing, where argument quality is prioritized. Furthermore, many consumers tend to stick to their comfort zones, relying on familiar recommendation patterns (P. Li and Tuzhilin, 2024). In such cases, user-based recommendations may disrupt this preference for familiarity and increase the cognitive effort required for evaluation (Kakaria et al., 2023). Consequently, this strategy may weaken argument quality and reduce the effectiveness of the recommendation under central processing.

The match of SGS and algorithmic recommendations fosters a "feel right" mentality, thereby enhancing the subsequent evaluation of the decision outcome (Khajehzadeh et al., 2014), manifesting as an elevated propensity to adopt. Hence, we hypothesize the following.

H1. For consumers with low (high) SGS, user-based (content-based) recommendations will be more effective in affecting consumers' adoption intention.

2.4. Perceived serendipity

Perceived serendipity describes the set of feelings that arise when a consumer encounters items or service that they did not directly choose (A. Kim et al., 2021). In recommendation systems, serendipity is defined as the degree to which recommendations are both unexpected and interesting to users (de Gemmis et al., 2015). It constitutes a pivotal component of the consumer experience, encompassing two distinct elements: unexpectedness and informational value (Grange et al., 2019). For consumers with low SGS, an inclination toward enjoying the browsing process has been observed. User-based recommendations have been found to generate a wider variety of unexpected yet equally valuable products (Herlocker et al., 2004), thereby increasing the likelihood of discovering serendipitous items (Adamopoulos and Tuzhilin, 2014). During the browsing process, consumers may encounter so-called "Ah-Hah" moments that motivate them to seek further information about the products (Niu et al., 2021). Consumers may recognize their need or desire for products they encounter while browsing (Sarantopoulos et al., 2019), leading to a greater willingness to adopt the recommendation. Unlike recommendations that solely align with previous behavior and stated preferences, user-based recommendations that introduce an element of chance tend to make consumers more satisfied (A. Kim et al., 2021). In accordance with the IAM, this serendipitous discovery allows consumers to perceive the value of the recommendation (Bao and Zhu, 2025), thereby increasing the likelihood of adoption. Hence, we suggest the following.

H2. Perceived serendipity mediates the relationship between the matching effect of user-based recommendations and low SGS on adoption intention.

2.5. Perceived diagnosticity

Perceived diagnosticity describes the extent to which algorithmic recommendations assist shoppers in accurately assessing products that match their preferences (Yi et al., 2017). The concept of perceived diagnosticity encompasses two fundamental components: goal-directedness and a principle of cognitive economy (B. Q. Liu and Goodhue, 2012). For consumers with high SGS, an inclination toward comparing target items with alternatives has been observed. Content-based recommendations present consumers with products that closely align with their needs, enabling them to select the most suitable option (Zhang et al., 2022) and achieve their goals with minimal cognitive effort. As a result, content-based recommendations enhance consumers' understanding of available alternatives (Yi et al., 2017), a feature that user-based recommendations do not provide. The existence of alternatives facilitates the process of comparison among consumers, enabling them to make an informed decision (Niu et al., 2021). By

fulfilling these two crucial aspects, consumers consider content-based recommendations to be more diagnostic. The perceived diagnosticity of a product has been shown to positively contribute to the cognitive evaluation of its attributes, thereby enhancing consumer adoption of the recommendations (Z. H. Jiang and Benbasat, 2004; Yang et al., 2023a). Hence, we suggest the following.

H3. Perceived diagnosticity mediates the relationship between the matching effect of content-based recommendations and high SGS on adoption intention.

2.6. Shopping motivation

Consumers' shopping motives, in addition to their shopping goals, are significant factors in understanding their shopping behavior. Previous studies have classified shopping motives into hedonic and utilitarian categories (Babin et al., 1994). Utilitarian motives are caused by the need to efficiently achieve a specific shopping goal, emphasizing goal orientation, efficiency, deliberation, and rational decision-making (S. Lee et al., 2025). Conversely, hedonic motivation propels consumers to seek a pleasurable browsing experience, characterized by entertainment, high involvement, and sensory arousal (S. Kumar and Sadarangani, 2021). Hedonic-motivated consumers have been observed to engage in exploratory information search (S. Kim and Eastin, 2011), while utilitarian-motivated consumers tend to seek information directly related to the products that align with their interest (Chiou and Ting, 2011).

Utilitarian motivation is a driving force that prompts consumers to prioritize the efficient and timely achievement of their shopping goals (A. Kumar and Kashyap, 2018), strengthening the central path of information processing. For consumers with high SGS, the adoption of recommendations is primarily influenced by argument quality. As a result, content-based recommendations, which precisely match functional needs, are likely to significantly outperform user-based recommendations. When consumers are hedonic-motivated, their inclination to adopt is more influenced by variety or emotional appeal (Lunardo and Mbengue, 2009). For consumers with low SGS, hedonic motivation fosters playful enjoyment during the shopping experience (Jantararat and Shannon, 2016), leading them to rely more on peripheral information processing paths. Bharadwaj and Bezborah (2021) reveal that extrinsic cues such as freebies and value framing significantly elevate impulsiveness, especially in scenarios where consumers lack a defined shopping goal (Bharadwaj and Bezborah, 2021). This lends direct support to our assertion that visually rich, algorithmic recommendation cues exert heightened influence when goal specificity is low, activating emotion-led decision pathways. Consequently, the efficacy of user-based recommendations is further amplified. Hence, we suggest the following.

H4. The interaction between shopping goal specificity and algorithmic recommendations is moderated by shopping motivation.

H4a. For consumers driven by utilitarian motivation, the matching advantage of content-based recommendations and high SGS is reinforced.

H4b. For consumers driven by hedonic motivation, the matching advantage of user-based recommendations and low SGS is reinforced.

3. Research methodology

3.1. Overview of the research

We focused on the interaction effect between algorithmic recommendations and shopping goal specificity and conducted three studies to examine our hypotheses. G*Power software indicated a required sample size of 210 for each study, setting effect size $f = 0.25$, $\alpha = 0.05$, and power = 0.95. To be on the safe side, we recruited more subjects for each study to ensure the processed data would meet the required statistical

power. Study 1 assesses the effect of matching algorithmic recommendations with shopping goal specificity (H1). Study 2 demonstrates the underlying mechanism (perceived diagnosticity and perceived serendipity) behind the matching effect (H2 and H3). Study 3 offers further support for the mechanism and examines the moderated effect of consumers' shopping motivation (H4, H4a, and H4b).

3.2. Study 1

Study 1 aims to test how the match between algorithmic recommendations (content-based vs. user-based) and shopping goal specificity (high vs. low) influences consumers' adoption intentions.

3.2.1. Method

Two hundred and forty volunteers from the wenjuanxing (www.wjx.cn) participated in return for ¥3. Wenjuanxing is an online sample marketplace in China and has been used by numerous scholars in the course of their research (Q. Chen et al., 2023; Peng et al., 2019). This study used a 2 (algorithmic recommendations: content-based vs. user-based) $\times$ 2 (shopping goal specificity: high vs. low) between-subjects design. Participants were randomly assigned to one of the 4 groups. During our research, we set up an attention check question about their reasons for seeing the recommended products: (1) content-based framework (e.g., my shopping history), (2) user-based framework (e.g., users similar to me like), (3) both, or (4) neither (Gai and Klesse, 2019). 23 participants who failed to correctly answer the reasons for the recommendations they saw were excluded from the analysis in each group. A total of 217 valid questionnaires ($M_{\text{age}} = 32.19$; Male = 41.9 %) were remained for further analysis. Our sample composition is highly related to the actual situation of online shopping in China, as published by CNNIC (CNNIC, 2024).

Firstly, the volunteers were given a segment of manipulation material concerning goal specificity and were subsequently instructed to complete a scale of shopping goal specificity (1 = *not at all specific*, 7 = *very specific*). In the low SGS group, participants were informed that "During their leisure time, they engaged in online shopping by accessing mobile apps. Their goal was to browse around, and they might buy some products." In the high SGS group, participants were informed that "During their leisure time, they engaged in online shopping by accessing mobile apps. Their goal was to look at potato chips and choose one" (Y. Kim and McFarland, 2024). Then, participants were asked to read the materials, including the recommendations and basic product information. We show 20 items on the interface for each condition, as evidenced by research findings that suggest this number does not result in excessive information overload (Willemsen et al., 2016). Participants in content-based recommendation were shown "Based on your choices, you might be interested in ...", while those in user-based recommendation were shown "Other users with similar interests also enjoyed ..." (Liao and Sundar, 2022). (See Appendix A for the material used). Once participants expressed their adoption intentions ($\alpha = 0.78$), they were also asked to fill in demographic information.

3.2.2. Results

Manipulation check. Results showed that participants in the high SGS group ($M_{\text{high}} = 5.32$) had significantly higher scores compared to those in the low SGS group ($M_{\text{low}} = 3.57$, $F(1, 215) = 110.29$, $p < 0.01$). Therefore, the manipulation of shopping goal specificity achieved the desired effect.

Adoption intentions. As algorithmic recommendations and shopping goal specificity are classification variables, one-way ANOVA is performed to test the moderation effect (W. Li et al., 2022). We conducted an ANOVA with algorithmic recommendations and shopping goal specificity as the independent variables and adoption intentions as dependent variable. The effect of algorithmic recommendations ($M_{\text{user}} = 5.18$, $SD = 1.01$; $M_{\text{content}} = 5.2$, $SD = 0.91$; $F(1, 215) = 0.018$, $p = 0.893$) and shopping goal specificity ($M_{\text{high}} = 5.13$, $SD = 1.07$; $M_{\text{low}} =$

5.25, $SD = 0.84$; $F(1, 215) = 0.895$, $p = 0.345$) on adoption intention were both not significant. This finding indicates that the methods employed for making recommendations and shopping goal specificity may not directly impact consumer acceptance. It aligns with the extant literature and the theoretical framework of this study, which posits that the efficacy of recommendation methods is also contingent on contextual factors. The interaction of algorithmic recommendations and shopping goal specificity was significant ($F(3, 213) = 3.664$, $p < 0.05$; see Fig. 2). For consumers with low SGS, participants in the user-based recommendation ($M_{\text{user}} = 5.44$, $SD = 0.84$) reported significantly higher adoption intentions than those in the content-based recommendation ($M_{\text{content}} = 5.06$, $SD = 0.81$; $F(1, 110) = 5.92$, $p < 0.01$, $\text{partial-}\eta^2 = 0.05$). For consumers with high SGS, participants in the content-based recommendation ($M_{\text{user}} = 4.92$, $SD = 1.1$) reported significantly higher adoption intentions than those in the user-based recommendation ($M_{\text{content}} = 5.35$, $SD = 1.0$; $F(1, 103) = 4.42$, $p < 0.05$, $\text{partial-}\eta^2 = 0.04$). Thus, H1 received confirmation.

The results provide support for the main hypothesis: the match between algorithmic recommendations and SGS positively impacted consumers' adoption intentions. Subsequently, an examination will be conducted of the mechanism under the matching effect (H2 and H3).

3.3. Study 2

Study 2 aims to test the mechanism behind the matching effect found in Study 1. The recommendation interface utilized in study 1 encompassed a diverse array of products, which introduced potential variations in outcomes due to factors such as price and the appeal of the product itself. In study 2, we narrowed down the product category to shoes with the objective of controlling for these factors to the greatest extent possible. This decision offered an additional advantage: it resulted in a different level of goal specificity compared to study 1.

3.3.1. Method

Study 2 used a 2 (algorithmic recommendations: content-based vs. user-based) $\times$ 2 (shopping goal specificity: high vs. low) between-subjects design (See Appendix B for the material). A total of two hundred and forty participants from China were recruited via Wenjuanxing, with each receiving ¥5 for their participation. They were randomly assigned to one of the above 4 conditions. Following the same procedure as study 1 (Gai and Klesse, 2019), a total of 227 valid questionnaires ($M_{\text{age}} = 32.15$; Male = 46.3 %) remained for further analysis.

The experimental procedure was analogous to that of Study 1, with the consumer shopping goals modified to 'sneakers (high SGS)' and 'shoes (low SGS)'. First, they were asked to fill in the scale of shopping goal specificity (1 = *not at all specific*, 7 = *very specific*). Participants were then shown an image of the recommendation interface. Participants in

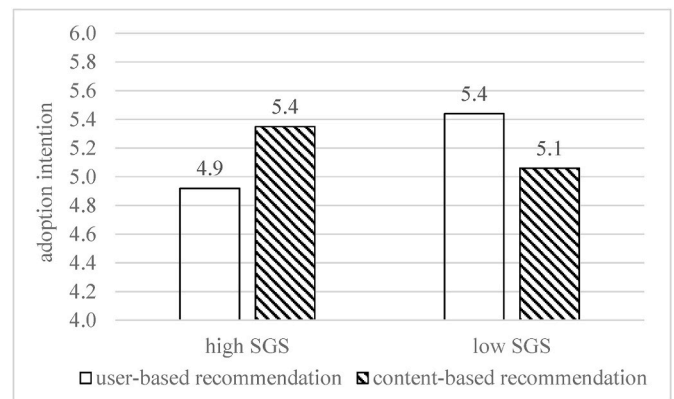


Fig. 2. The interaction between algorithmic recommendations and shopping goal specificity in predicting adoption intentions (Study 1).

the content-based recommendations were told that “guess your favorite products”. Participants in the user-based recommendations were told that “products liked by users similar to you (Zhang et al., 2022).” Participants were required to report their adoption intentions ($\alpha = 0.79$) (Bao and Zhu, 2025), perceived diagnosticity ($\alpha = 0.83$), and perceived serendipity ($\alpha = 0.82$) (Yi et al., 2017) after viewing the materials. For caution, we measured shopping habits (“How often do you shop online?”; 1 = Never, 7 = Very frequently) and website preference (“How much do you like this website?”; 1 = Do not like at all, 7 = Like very much) as confounding variables (Xu et al., 2023). Finally, participants provided their demographic information.

3.3.2. Results

Reliability test. The result from SPSS 25 indicated that Cronbach’s α for adoption intention, perceived diagnosticity, perceived serendipity was 0.79, 0.83, 0.82. CR values were 0.830, 0.865, 0.867, higher than 0.7. AVE values were 0.619, 0.616, 0.619, higher than 0.5. Harman’s single-factor test shows that no significant common factor was extracted, and the main factor explained 39.73 % of the variance, lower than the threshold value of 50 % (Podsakoff and Organ, 1986). This indicates that the common method bias issue is not significant. Then, we tested the data for normal distribution. The absolute values of the skewness and kurtosis coefficients ranged from 0.602 to 0.922 and from 0.115 to 1.186, respectively. According to the criteria proposed by Kline (2011) - absolute values of skewness less than 3 and absolute values of kurtosis less than 10 (Kline, 2011) - our data exhibited characteristics of a normal distribution. All VIF values are between 1 and 2, and all correlations are less than 0.7 (Dormann et al., 2013). Therefore, we believe there is no multicollinearity issue.

Manipulation check. The results showed that participants in the high SGS condition ($M_{\text{high}} = 5.24$) had significantly higher scores compared to those in the low SGS condition ($M_{\text{low}} = 3.85$, $F(1, 225) = 65.17$, $p < 0.01$). Thus, the shopping goal specificity manipulation achieved the desired effect.

Adoption intentions. As algorithmic recommendations and shopping goal specificity are classification variables, one-way ANOVA is performed to test the main effect (W. Li et al., 2022). We conducted an ANOVA with algorithmic recommendations and shopping goal specificity as the independent variables and adoption intentions as the dependent variable. The effect of algorithmic recommendations ($M_{\text{user}} = 4.95$, $SD = 1.03$; $M_{\text{content}} = 5.06$, $SD = 0.87$; $F(1, 225) = 0.782$, $p = 0.38$) and shopping goal specificity ($M_{\text{high}} = 5.01$, $SD = 1.07$; $M_{\text{low}} = 5.00$, $SD = 0.82$; $F(1, 225) = 0.001$, $p = 0.98$) were both not significant. The interaction effect of algorithmic recommendations and shopping goal specificity was significant ($F(3, 223) = 6.71$, $p < 0.01$; see Fig. 3). As found in Study 1, for consumers with low SGS, participants in the user-based recommendation reported significantly higher adoption

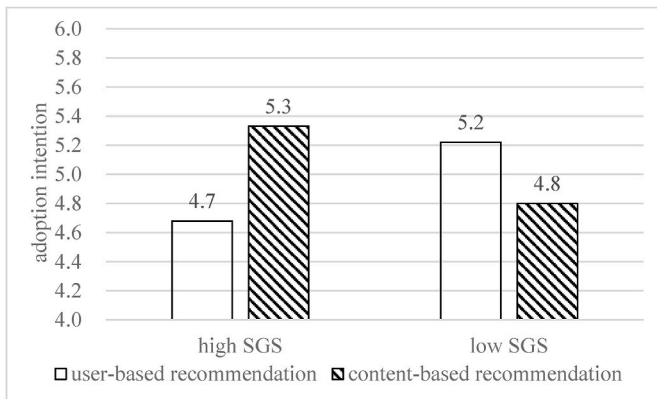


Fig. 3. The interaction between algorithmic recommendations and shopping goal specificity in predicting adoption intentions (Study 2).

intentions than those in the content-based recommendation ($M_{\text{user}} = 5.22$, $SD = 0.79$; $M_{\text{content}} = 4.8$, $SD = 0.80$; $F(1, 110) = 7.94$, $p < 0.01$, $\text{partial-}\eta^2 = 0.07$). For consumers with high SGS, participants in the content-based recommendation reported significantly higher adoption intentions than those in the user-based recommendation ($M_{\text{user}} = 4.68$, $SD = 1.17$; $M_{\text{content}} = 5.33$, $SD = 0.86$; $F(1, 112) = 11.22$, $p < 0.01$, $\text{partial-}\eta^2 = 0.09$). Thus, H1 received additional confirmation.

Mediation. First, we examined the main effect of algorithmic recommendations * SGS on perceived diagnosticity ($M_{\text{content}} = 5.43$, $SD = 0.80$; $M_{\text{user}} = 4.72$, $SD = 1.18$; $F(1, 112) = 14.27$, $p < 0.01$) in high SGS and perceived serendipity ($M_{\text{user}} = 5.41$, $SD = 0.81$; $M_{\text{content}} = 4.82$, $SD = 0.92$; $F(1, 110) = 13.15$, $p < 0.01$) in low SGS. To further strengthen the support for our proposed explanation, we employed Model 4 in Process - a macro for SPSS (Hayes, 2013) - with 5000 bootstrap samples (see Figs. 4 and 5 for more details). As predicted, mediation through perceived diagnosticity ($\beta = 0.42$, $SE = 0.13$, 95 % CI [0.192, 0.685]) to adoption intention was significant, and perceived serendipity ($\beta = 0.04$, $SE = 0.05$, 95 % CI [-0.151, 0.055]) to adoption intention was not significant in high SGS. Also, mediation through perceived serendipity ($\beta = 0.24$, $SE = 0.09$, 95 % CI [0.073, 0.425]) to adoption intention was significant, and perceived diagnosticity ($\beta = -0.01$, $SE = 0.02$, 95 % CI [-0.049, 0.023]) to adoption intention was not significant in low SGS. Thus, H2 and H3 were supported.

Control variables and alternative explanations. The interaction between algorithmic recommendations and shopping goal specificity remained significant after controlling for age and gender ($p < 0.01$). The effect of participants’ habits and website preferences on adoption intention was not significant ($p > 0.10$).

The findings of Study 2 demonstrate a significant matching effect (H1). Additionally, the findings indicate that perceived diagnosticity and perceived serendipity mediate this effect (H2, H3).

3.4. Study 3

Study 3 aims to retest the mediation of perceived serendipity and diagnosticity and examine the moderated effect of consumers’ shopping motivation (H4, H4a, and H4b). In study 3, we further eliminated potential confounding factors by presenting the participants with the same shopping interface.

3.4.1. Method

A total of 256 volunteers ($M_{\text{age}} = 32.7$; Male = 33.2 %) from Wenjuanxing participated in return for ¥5 (see Appendix E for details of all studies). Our sample composition is highly related to the actual situation of online shopping in China, as published by CNNIC (CNNIC, 2024). Study 3 used a 2 (interaction: content-based*high SGS vs. user-based*low SGS) by 2 (shopping motivation: utilitarian vs. hedonic) between-subjects design (Y. Kim and McFarland, 2024) (See Appendix C for the material). Participants were randomly assigned to one of the 4 groups. Participants in the hedonic motivation group were informed that “This smartwatch features a sleek, modern design and offers a variety of interactive smart applications, delivering a playful and enjoyable user

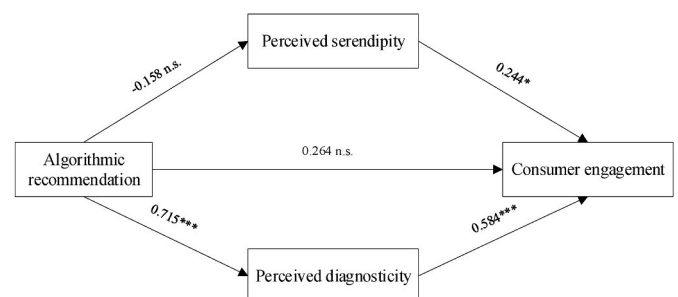


Fig. 4. Result of mediation analysis of high SGS condition (Study 2).

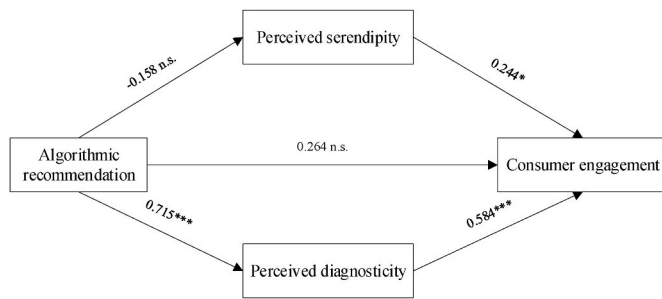


Figure 4. Result of mediation analysis of high SGS condition (Study 2)

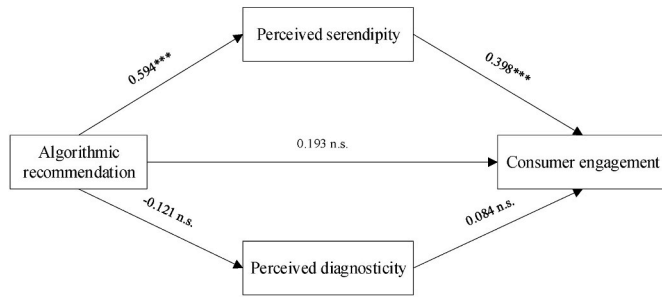


Fig. 5. Result of mediation analysis of low SGS condition (Study 2).

experience. Its aesthetic appeal and entertaining functionalities make it particularly attractive to users seeking pleasure and enjoyment during use.” Participants in the utilitarian motivation group were informed that “This smartwatch is equipped with a high-definition square or round display and supports a wide range of practical functions, including sports tracking and health monitoring. Its long battery life and functional design make it a reliable tool for users pursuing efficiency, functionality, and everyday utility.” And then they answered “when you thought about purchasing a smartwatch, consider using the money you received for something enjoyable or something practical?” (1 = *Something enjoyable*, 7 = *Something practical*) (S. Chen et al., 2021). Participants were required to report adoption intention ($\alpha = 0.74$), perceived diagnosticity ($\alpha = 0.76$), perceived serendipity ($\alpha = 0.80$) (See Appendix D for more details) after viewing the material. Finally, participants provided their demographic information.

3.4.2. Results

Reliability test. The result from SPSS 25 indicated that Cronbach’s α for adoption intention, perceived diagnosticity, perceived serendipity was 0.74, 0.76, 0.80. CR values were 0.818, 0.837, 0.857, higher than 0.7. AVE values were 0.6, 0.562, 0.599, higher than 0.5. Harman’s single-factor test shows that no significant common factor was extracted, and the main factor explained 30.5 % of the variance, lower than the threshold value of 50 % (Podsakoff and Organ, 1986). This indicates that the common method bias issue is not significant. Then, we tested the data for normal distribution. The absolute values of the skewness and kurtosis coefficients ranged from 0.494 to 1.052 and from 1.177 to 2.821, respectively. According to the criteria proposed by Kline (2011) - absolute values of skewness less than 3 and absolute values of kurtosis less than 10 (Kline, 2011) - our data exhibited characteristics of a normal distribution. All VIF values are between 1 and 2, and all correlations are less than 0.7 (Dormann et al., 2013). Therefore, we believe there is no multicollinearity issue. See Appendix F for details.

Manipulation check. The results indicated that participants in the hedonic motivation condition ($M_{\text{hedonic}} = 4.41$) had significantly higher scores compared to those in the utilitarian condition ($M_{\text{utilitarian}} = 2.6$, $F(1, 254) = 122.56$, $p < 0.01$). Thus, the shopping motivation manipulation achieved the desired effect.

Adoption intentions. We conducted a two-way ANOVA with algorithmic recommendations and shopping goal specificity as the

independent variables and adoption intentions as the dependent variable. The effect of algorithmic recommendations ($F(1, 252) = 0.3$, $p = 0.58$) and shopping goal specificity ($F(1, 252) = 1.21$, $p = 0.27$) was not significant. The interaction effect of algorithmic recommendations and shopping goal specificity was significant ($F(1, 252) = 55.61$, $p < 0.01$; see Fig. 6). For consumers with hedonic motivation, participants in the user-based recommendation*low SGS reported significantly higher adoption intention than those in the content-based recommendation*high SGS ($M_{\text{user}} = 5.6$, $SD = 0.84$; $M_{\text{content}} = 5.02$, $SD = 0.84$; $F(1, 126) = 23.78$, $p < 0.01$, $\text{partial-}\eta^2 = 0.16$). For consumers with utilitarian motivation, participants in the content-based recommendation*high SGS reported significantly higher adoption intention than those in the user-based recommendation*low SGS ($M_{\text{user}} = 5.06$, $SD = 0.81$; $M_{\text{content}} = 5.74$, $SD = 0.51$; $F(1, 126) = 32.16$, $p < 0.01$, $\text{partial-}\eta^2 = 0.20$). Thus, H4 received confirmation.

Mediation analysis for perceived diagnosticity. To further strengthen the support for our proposed hypotheses, we employed Model 8 in Process - a macro for SPSS (Hayes, 2013) - with 5000 bootstrap samples. We conducted with match strategy as the independent variable, shopping motivation as the moderator, perceived diagnosticity as the mediator, and adoption intention as the dependent variable. As predicted, the moderated mediation effect reached significance (index = 0.193, $SE = 0.098$, 95 % $CI = [0.02, 0.40]$). In the utilitarian motivation condition, perceived diagnosticity was a significant mediator (indirect effect = 0.19, $SE = 0.09$, 95 % $CI = [0.02, 0.37]$). However, in the hedonic motivation, perceived diagnosticity is not a significant mediator (indirect effect = -0.003 , $SE = 0.03$, 95 % $CI = [-0.06, 0.05]$).

Mediation analysis for perceived serendipity. To further strengthen the support for our proposed hypotheses, we employed Model 8 in Process with 5000 bootstrap samples. We conducted with match strategy as the predictor, shopping motivation as the moderator, perceived diagnosticity as the mediator, and adoption intention as the dependent variable. The results are shown in Table 1. As predicted, the moderated mediation effect reached significance (index = 0.168, $SE = 0.08$, 95 % $CI = [0.03, 0.34]$). In the hedonic motivation condition, perceived serendipity was a significant mediator (indirect effect = -0.19 , $SE = 0.06$, 95 % $CI = [-0.32, -0.08]$). However, in the utilitarian motivation condition, perceived serendipity was not a significant mediator (indirect effect = -0.02 , $SE = 0.05$, 95 % $CI = [-0.11, 0.08]$). The results are shown in Table 2.

Study 3 further supports the crucial role that perceived serendipity and diagnosticity play in the matching effect. Study 3 also assessed shopping motivation (hedonic vs. utilitarian) as a theoretically relevant moderating factor. The matching of content-based recommendation and high SGS (user-based recommendation and low SGS) increased consumers’ perceived diagnosticity (serendipity), and thus, their adoption intentions were enhanced, which is stronger among utilitarian (hedonic)

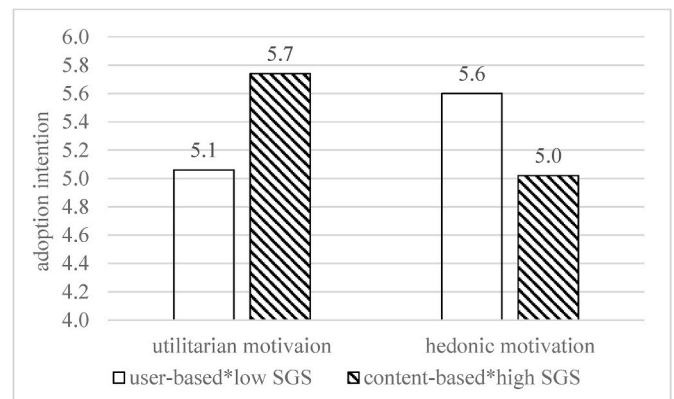


Fig. 6. The interaction between matching strategy and shopping motivation in predicting adoption intentions (Study 3).

Table 1
Regression results of model 8 (Study 3).

Variables	perceived serendipity	perceived diagnosticity	adoption intention
matching strategy	−0.5***	−0.016	−0.397***
shopping motivation	−0.484***	0.008	−0.361**
perceived serendipity			0.377***
perceived diagnosticity			0.220***
matching strategy*shopping motivation	0.445**	0.875***	0.91***
cons	5.379***	5.125***	2.449***
R ²	0.093***	0.262***	0.35***

Note. **, *** separately represents significant at the 1 %, and 1 % level.

Table 2
Moderated mediation analysis (Study 3).

	effect	β	SE	LLCI	ULCI
perceived serendipity	hedonic	−0.189	0.061	−0.323	−0.083
	utilitarian	−0.021	0.048	−0.11	0.084
	index	0.168	0.08	0.035	0.344
perceived diagnosticity	hedonic	−0.003	0.026	−0.062	0.046
	utilitarian	0.189	0.091	0.02	0.375
	index	0.193	0.098	0.02	0.404

motivated consumers.

4. General discussion

The findings from the three studies offer converging evidence that lends support to our theoretical framework. Content-based recommendations for consumers with high SGS increased perceived diagnosticity, ultimately leading to higher adoption intentions. User-based recommendations for consumers with low SGS increased perceived serendipity, ultimately leading to higher adoption intentions. Across three studies, we first demonstrated the positive effect of matching algorithmic recommendations with shopping goal specificity and our proposed underlying mechanism. Shopping motivation (hedonic and utilitarian) is a significant moderator of the matching effect.

In Study 1, the matching effect between algorithmic recommendations and shopping goal specificity was validated. This finding differs from previous research but aligns with analogous underlying principles. As indicated by prior research, during the initial stage of the shopping process (which, in this study, corresponds to low SGS), consumers express a desire to consider a greater number of products (Song et al., 2017). In the second stage (which, in this study, corresponds to high SGS), consumers express a desire to evaluate target products in an effective manner (Yan et al., 2016). This finding aligns with the central tenet of our study, suggesting a correlation between high (low) SGS and content-based (user-based) recommendations.

In Study 2, we validated the mediating effects of perceived serendipity and perceived diagnosticity (Grange et al., 2019; Niu et al., 2021). This finding aligns with previous research on IAM (Yang et al., 2023a, 2023b; Yi et al., 2017), which has indicated their role in consumer persuasion. However, our study differs in that we distinguished between their mediating pathways.

In Study 3, the moderating role of shopping motivation was validated. Shopping motivation provides a valuable perspective for research in algorithmic recommendations (Babin et al., 1994). Individuals driven by hedonic motivations exhibit heightened sensitivity to serendipity, seeking out novel experiences and displaying a preference for a variety of products (Lunardo and Mbengue, 2009). Conversely, consumers with a utilitarian motivation exhibit heightened sensitivity to diagnosticity, seeking to efficiently accomplish their shopping goals (A. Kumar and Kashyap, 2018).

4.1. Theoretical implications

These findings allow us to make several theoretical contributions. First, by examining shopping goal specificity exhibited by consumers during the shopping apps usage, this study seeks to further advance the comprehension shopping goal theory. Past research has centered on the utilization of disparate marketing strategies for different shopping stages (Luo et al., 2019; Song et al., 2017; Virdi et al., 2020; Wan et al., 2024; Yan et al., 2016). This study demonstrates that, in addition to considering milestones, immediate goals are also an important factor influencing consumers adopt intentions. This finding emphasizes the influence of real-time contextual factors on the efficacy of recommendations.

Second, we propose a matching effect that will promote information adoption by integrating shopping goals theory and information adoption models. Shopping goals theory posits that consumers have disparate shopping goals at different shopping stages (L. Lee and Ariely, 2006; Wan et al., 2024). However, there is no research on how to set recommendation strategies based on shopping goals - the “if-then” guidance framework (Y. Kim and McFarland, 2024). This research suggests that the matching of algorithmic recommendations based on shopping goal specificity will positively influence consumers’ adoption intention. Specifically, for consumers with low (high) SGS, user-based (content-based) recommendations will be more effective in affecting consumers adoption intention. Despite the existence of prior studies that have proposed research frameworks such as regulatory-fit (Zhang et al., 2024) and exploration-exploitation (Bastani et al., 2022), it is evident that our shopping goal specificity differs from theirs. Two factors underpin this phenomenon. On the one hand, shopping goals are dynamic and change throughout the consumer’s shopping journey. In contrast, regulatory focus (whether consumer traits, product features, or content frameworks) is generally stable. This distinction contributes to the alignment of our matching strategy with real-world scenarios. On the other hand, exploration-exploitation entails a balanced exploitation of prior information and the exploration of novel information, with a primary objective being the optimization of algorithms to achieve this equilibrium (Tehrani and Ching, 2024). This study elucidates a straightforward trade-off logic, which posits the utilization of shopping goal specificity to alternate between disparate recommendation algorithms. Therefore, our matching effect serves to expand the boundary of shopping goal theory and deepen the understanding of the relative effectiveness of algorithmic recommendations.

Third, we offer robust support for the underlying mechanism (perceived diagnosticity and perceived serendipity) based on the information adoption model (Sussman and Siegal, 2003). It explores the various factors that influence perceived usefulness (Shen et al., 2016; Y. Sun et al., 2019; Venciute et al., 2023), which will then influence adoption intention. The term “serendipity” has been defined as “finding something valuable or consensual without looking for it (A. Kim et al., 2021).” Hence, we propose that perceived serendipity can reflect perceived usefulness to some extent in this research. Furthermore, research also proposes that perceived diagnosticity is understood to signify one’s perceived usefulness of the information (Bao and Zhu, 2025). While prior work acknowledges the roles of diagnosticity and serendipity, it conflates their mediating pathways within a monolithic framework (Yang et al., 2023a, 2023b; Yi et al., 2017). Departing from this tradition, we propose a dual-path mediation model: perceived diagnosticity (serendipity) mediates the matching effect of content-based recommendations and high SGS (user-based recommendations and low SGS).

Finally, we propose a subsequent extension of the IAM, indicating the impact of shopping motivation on the information processing pathway. Previous studies (Babin et al., 1994) have classified shopping motives into hedonic and utilitarian. We conceptualize shopping motivation as a moderator that dynamically allocates cognitive resources between central and peripheral processing routes. Utilitarian motivation

(focus more on tasks; S. Chen et al., 2021) amplifies the central path, intensifying the mediating role of perceived diagnosticity in content-based recommendation adoption (for consumers with low SGS). Hedonic motivation (focus more on experience; A. Li et al., 2024) triggers the peripheral path, enhancing the mediating role of perceived serendipity in user-based recommendation adoption (for consumers with high SGS). This “goal-motivation-algorithm” congruence framework offers a more granular lens for human-algorithm interaction research.

4.2. Managerial implications

The results of our research suggest that a variety of algorithmic recommendations must be tailored to align with the distinct goals of consumers when using shopping apps. Despite the inability to directly quantify the ROI of algorithmic adaptive switching, extant research reports (Abraham et al., 2024) indicate that personalized product recommendations can engender a multitude of benefits (e.g., 10–20 % of cross-selling for multi-category retailers, a 50 % reduction in cart abandonment rates, and a 10 % or more increase in sales of perishable goods), implying the application potential of our findings. Consequently, managers must prioritize not only the optimization of recommendation algorithms but also the identification of consumers’ needs in terms of goal, motivation, and so on. Thus, our research yields several managerial implications as outlined below:

Firstly, while the provision of directly deployable algorithm code is not currently feasible, laboratory experiments have been conducted with strict manipulation, thereby eliminating confounding factors. Managers must develop refined consumer profiles (e.g., distinguishing between aimless and purposeful shopping) by analyzing behavioral patterns, shopping frequency, search behaviors, and clickstream data. The most recent research findings by Ding et al. (2025) provide effective insights into the classification of consumer shopping goal specificity (Ding et al., 2025). They use sequence analysis and cluster analysis to analyze the sources of consumer clicks on products, distinguishing three types of browsing patterns with varying degrees of shopping goal specificity (precise keywords, general categories, and external sources). Based on this classification, recommendation strategies should be aligned accordingly. When consumers are identified as performing quick keyword searches or consecutive keyword searches, they should be defined as high SGS users. At this point, the system will use content-based recommendations (i.e., “recommend products according to your search history”). When consumers are identified as performing external triggered visits, they should be defined as low SGS users. At this point, the system will switch to user-based recommendations (i.e., “recommend products favored by someone similar to you”).

Second, in the context of recommendation adoption, serendipity and diagnosticity emerge as critical psychological drivers. Therefore, managers should not only provide accurate recommendations but also introduce occasional surprising elements. For example, users who engage with shopping platforms across multiple contexts (e.g., commuting, leisure time, late-night browsing) tend to seek a broader range of information (Wang et al., 2022). In addition, the variety-seeking framework, containing three key components (P. Li and Tuzhilin, 2024), can help determine when and how to introduce novel content. In such cases, inserting novel items to maintain interest and stimulate discovery is necessary. On the other hand, the incorporation of a product attribute comparison function (e.g., information on price, score, material, etc.) may empower consumers to make more informed decisions between different products, thereby facilitating their decision-making process.

Finally, consumers’ likelihood of adopting recommendations is significantly influenced by their underlying shopping motivations. Thus, it is essential for managers to tailor recommendation strategies to accommodate both utilitarian and hedonic motivations. For instance, women are more likely to exhibit hedonic motivations, while men tend

to show stronger utilitarian tendencies (Diep and Sweeney, 2008). In utilitarian contexts, recommendations should emphasize functional attributes (e.g., expert reviews, energy efficiency, or durability) to support rational decision-making. In contrast, for hedonic consumers, emotionally engaging content (e.g., mystery bundles or unlocking new uses of XX) can enhance enjoyment and deepen involvement.

4.3. Limitations and future research

First, Kim et al. (2025) advocate for research in Asian countries, where emerging technologies, such as digital engagement, are particularly prevalent (J. Kim et al., 2025). Therefore, our study, which is based on Chinese samples to avoid confounding factors (J. Kim, 2024), supplements previous research to a certain extent. Despite studies demonstrating an absence of discrepancy in algorithm aversion between collectivist countries (e.g., India) and individualist countries (e.g., the United States) at an aggregate level (N. T. Y. Liu et al., 2023), cultural differences in the acceptance of algorithmic recommendations have yet to be verified. Hence, future studies need samples from different cultures and demographics to improve the generalizability of our findings. It is possible that different cultural environments may affect recommendation effects (Lim et al., 2025). For instance, social identity tendencies in collectivist cultures (e.g., sensitivity to user-based recommendations) may amplify the experimental results. Conversely, the efficacy of content-based recommendations may be more pronounced in markets characterized by individualist culture.

Second, this study is grounded in a laboratory setting and boasts high internal validity. As Kim (2024) notes in his article, experiments are frequently employed in social science research and can offer compelling evidence of causal relationships between variables (J. Kim, 2024). However, due to the absence of comprehension regarding algorithms, the design of shopping pages recommended by diverse algorithms for our experimental procedures proves challenging. Furthermore, laboratory experiments hinder the ability to accurately quantify actual purchases, clicks, associated behaviors, and the cost of switching algorithms. In accordance with the theory of planned behavior, behavioral intention has been demonstrated to be a direct predictor of behavior (De Cannière et al., 2009). However, the existence of the “intention-behavior” gap necessitates further measurement of actual behavior. Future research should extend these findings by using behavioral measures such as real purchase decisions, lab or field experiments, and longitudinal tracking in natural settings. Additionally, studies could develop adaptive recommendation systems that adjust algorithms in real time based on shopping goal specificity. Combining behavioral validation with dynamic personalization can enhance both the practical impact and scientific value of these insights.

Third, the exploration of composite goal scenarios may deepen the understanding of consumers’ multifaceted goals during the shopping process. When consumers have both primary and secondary goals (e.g., primarily buying laptops and secondarily browsing for accessories), it would be more relevant to real shopping cart scenarios to investigate the synergistic effects of hybrid recommendation strategies (e.g., primary + long-tail recommendations) (X. Huang and Wu, 2019; Sreepada and Patra, 2021). For instance, when consumers seek to procure a computer, the additional recommendation of complementary products, may further enhance their inclination to purchase both the primary and additional items.

Finally, although this study did not address the issue, algorithmic ethics cannot be overlooked, particularly with regard to online retail platforms that rely on data operations. Consumers expect platforms to handle their data responsibly and become dissatisfied when their privacy is violated (He et al., 2025). Practices that enhance algorithmic transparency, such as providing post-hoc explanations, can strengthen consumer trust (Q. Q. Chen et al., 2025). Future research could explore ways to improve explainability (e.g., data sources and algorithmic principles) under different algorithmic recommendation methods.

Another area for future research is whether allowing consumers to choose which algorithm to use would lead to greater satisfaction.

CRedit authorship contribution statement

Jie Xin: Writing – review & editing, Validation, Software, Project administration, Funding acquisition, Conceptualization. **Peng Qin:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Tingting Li:** Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Zhihao Yang:** Writing – review & editing, Supervision, Methodology, Investigation.

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Declaration of competing interest

There are no conflicts of interest to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jretconser.2025.104509>.

Data availability

Data will be made available on request.

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